Coordinate Plane, Line Plots, and Numerical Patterns — Combined Practice

Answer Key

Grade 4 / Grade 5

Quick Answers

1. 40	4. 4	7. 4	10. 48
2. 18	5. 35	8. 6	11. 8
3. 2	6. —	9. 60	12. —

Common wrong answers

4. *If they got 8: divided the total by the number of different book counts on the plot instead of by the number of students*
5. *If they got 15: used the tallest stack on the plot (the most common number of books) instead of the largest number of books read*
9. *If they got 45: lined the two patterns up one step apart, pairing a term of the first pattern with the term before it in the second*
10. *If they got 12: counted the students instead of adding up the books they read*

How to read this key. Every problem here needs two pieces of information joined together, so each solution names the two pieces first and only then computes. The recurring student error is using one piece and guessing the other: reading a line plot but forgetting that each × is a student rather than a book, or pairing two patterns one step out of step. Both are shown below where they occur.

1. Two patterns, one ordered pair.

Both patterns start at zero, so matching terms are the 1st with the 1st, the 2nd with the 2nd, and so on. Pattern A grows by 2 a step and Pattern B by 8 a step, so after the same number of steps B has grown $8 \div 2 = 4$ times as much as A. Every ordered pair therefore satisfies $B = 4 \times A$.

$$B = \frac{8}{2} \times 10 = 4 \times 10$$

The ordered pair is $(10, 40)$, so the B-value is $\boxed{40}$.

2. Working the same relationship backwards.

The multiplier is the one found above: each step Pattern B gains 8 while Pattern A gains 2, so $B = 4 \times A$, i.e. the **multiplier is 4**.
Going from a B-value back to its A-value undoes that multiplication, so divide:

$$A = 72 \div \frac{8}{2} = 72 \div 4$$

The A-value is $\boxed{18}$, and the ordered pair is $(18, 72)$.
Check: $18 \times 4 = 72$ ✓

3. Reading one column of the line plot.

Find 5 on the number line and count the $\times$ marks stacked above it — each $\times$ is one student, not one book. There are two marks above 5.
So $\boxed{2}$ students read exactly 5 books, and as an ordered pair (books read, number of students) that column is $(5, 2)$.

4. The mean of the whole plot.

A mean needs two numbers: the total of all the values, and how many values there are. Read the total straight off the plot by multiplying each book count by the size of its stack, then add:

$$2(2) + 3(3) + 3(4) + 2(5) + 1(6) + 1(7) = 4 + 9 + 12 + 10 + 6 + 7 = 48$$

Twelve students are on the plot, so divide the total by the number of students:

$$\text{mean} = \frac{48}{12}$$

The mean is $\boxed{4}$ books per student.
Watch for: dividing by the number of *columns* on the plot instead of the number of students, which gives $48 \div 6 = 8$ — a mean twice as large as the biggest stack on the plot, and impossible.

5. A rule applied to a value read off the plot.

Two pieces: the plot supplies the number of books, the club rule supplies the points per book. The rightmost mark on the plot sits above 7, so the student who read the most books read 7 of them. The rule gives 5 points a book:

$$7 \times 5 = 35$$

That student earned $\boxed{35}$ points.
Watch for: using the *tallest stack* instead of the *rightmost* mark. The tallest stacks are at 3 and 4 books, which is where most students are — not where the biggest reader is. Reading 3 books would give only $3 \times 5 = 15$ points.

6. Two patterns, plotted.

List each pattern first, then pair matching terms:

- Pattern P: 1, 4, 7, 10 (start at 1, add 3)
- Pattern Q: 3, 12, 21, 30 (start at 3, add 9)

Ordered pairs: $(1, 3)$, $(4, 12)$, $(7, 21)$, $(10, 30)$. Plotted on Grid A those four points climb steeply and lie on a straight line through the origin.
The relationship in words: *each Q-value is 3 times its P-value.* Q grows 9 a step while P grows 3 a step, and $9 \div 3 = 3$, which is exactly the multiplier the plotted points show.
Plot-and-describe task: $\boxed{\text{open response}}$ — check the four plotted points and the wording of the description.

7. The median of the plot.

The median is the middle value when all twelve values are in order, so count × marks from the left rather than reading the middle of the number line:

$$2,\ 2,\ 3,\ 3,\ 3,\ \underline{4,\ 4},\ 4,\ 5,\ 5,\ 6,\ 7$$

With an even number of students the median is the mean of the 6th and 7th values, both 4, so the median is $\boxed{4}$ books.
It is **equal** to the mean found earlier. A plot whose mean and median agree is roughly balanced: the long thin tail out at 6 and 7 books is offset by the pile of students at 2 and 3.

8. Shifting every value by the same amount.

Every one of the twelve values goes up by the same 2 books, so the total goes up by $12 \times 2 = 24$ and the mean goes up by exactly 2 — the whole plot slides two spaces right without changing shape.

$$\text{new mean} = 4 + 2$$

The new mean is $\boxed{6}$ books.
That is the reason no re-adding is needed: adding a constant to every value adds the same constant to the mean. (The long way confirms it: the new total is $48 + 24 = 72$, and $72 \div 12 = 6$.)

9. Find Sara's mistake.

Pattern S: 0, 5, 10, 15, 20, ... Pattern T: 0, 15, 30, 45, 60, ...
T grows 15 a step while S grows 5 a step, so $T = 3 \times S$ for matching terms:

$$T = \frac{15}{5} \times 20 = 3 \times 20$$

The correct pair is $(20, 60)$, so the T-value is $\boxed{60}$.
Sara's mistake: she lined the two lists up one step out of step. The value 20 is the 5th term of Pattern S, but 45 is only the 4th term of Pattern T. Pairing term 5 with term 4 gives $(20, 45)$. Matching terms must sit in the *same position* in both patterns.

10. The class total.

Each × is one student, and the number under it is how many books that student read, so the total is a sum of stack-size times book-count:

$$2(2) + 3(3) + 3(4) + 2(5) + 1(6) + 1(7) = 4 + 9 + 12 + 10 + 6 + 7$$

The class read $\boxed{48}$ books.
As an ordered pair, (number of students, total books read) is $(12, 48)$.
Watch for: answering 12. That is the number of × marks — the students, not the books. The plot's marks count people; the labels underneath count books.

11. A rule applied to the class total.

The bonus rule needs the class total, which came from the plot: 48 books. Bonus points are awarded only for books *beyond* 40, so subtract rather than divide:

$$48 - 40 = 8$$

The class earns $\boxed{8}$ bonus points.

12. Challenge: a pattern built on top of the data.

Month one is the class total already found, 48 books, and each later month adds 6:

Month	1	2	3	4
Class total	48	54	60	66

As ordered pairs (month number, class total): $(1, 48)$, $(2, 54)$, $(3, 60)$, $(4, 66)$.
The pattern in words: the first number of each pair goes up by 1 while the second goes up by 6, so every step right on a grid is six steps up. Plotted, the four points would lie on a straight line climbing to the right — and, unlike the patterns in problems 1 and 9, that line would *not* pass through the origin, because month one already starts at 48 books rather than at zero.
Accept any description naming the constant increase of 6 and a straight line of points. Open synthesis task: $\boxed{\text{open response}}$ — check the table, the ordered pairs, and the description.